

Companion to Bird & de Moor, Algebra of Programming

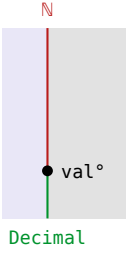
§6. Recursive Programs

§6.1. Digits of a number

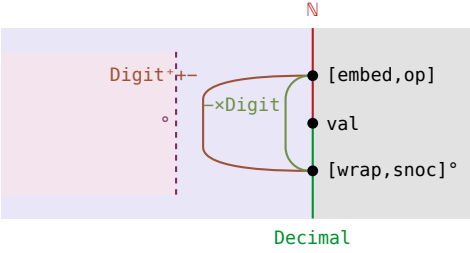
(6.1a)

$val^\circ = embed^\circ wrap \cup op^\circ -xDigit(val^\circ)snoc$

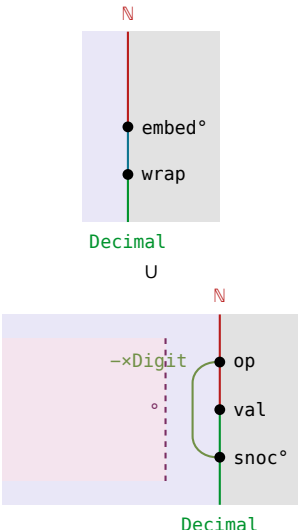
a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest



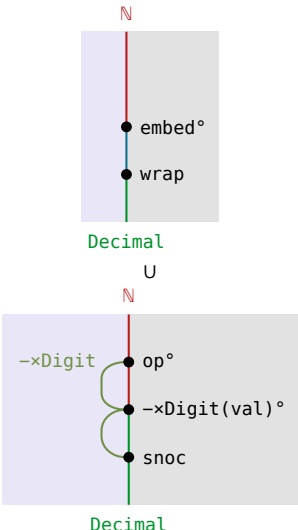
(a)



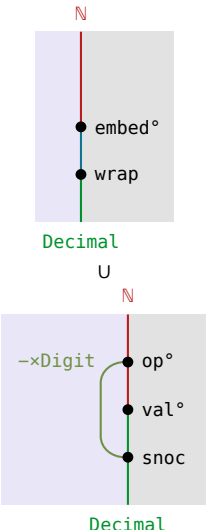
(b)



(c)



(d)



(e)

(a)	val°	the converse of val , which reads digits as a number
(b)	$([wrap,snoc]^\circ Digit^{++} - (-xDigit(val)) [embed,op])^\circ$	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(c)	$(wrap^\circ embed)^\circ \cup (snoc^\circ -xDigit(val) op)^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(d)	$embed^\circ wrap \cup op^\circ -xDigit(val)^\circ snoc$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(e)	$embed^\circ wrap \cup op^\circ -xDigit(val^\circ) snoc$	undoing the read of the front while keeping the last digit is val° on the front alone

$op^\circ(m,p) \iff \pi_1(p)=m/10 \wedge \pi_2(p)=m \% 10$ op° splits a number into its quotient and remainder by 10 $(\exists p. 0 < \pi_1 p \wedge op^\circ(m,p)) \iff 10 \leq m$ op° gives a pair with a nonzero first component exactly at the numbers with two or more digits $(\exists d. embed^\circ(m,d)) \iff 0 < m \wedge m < 10$ $embed^\circ$ gives a digit exactly at the one-digit numbers	(6.1b)
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§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

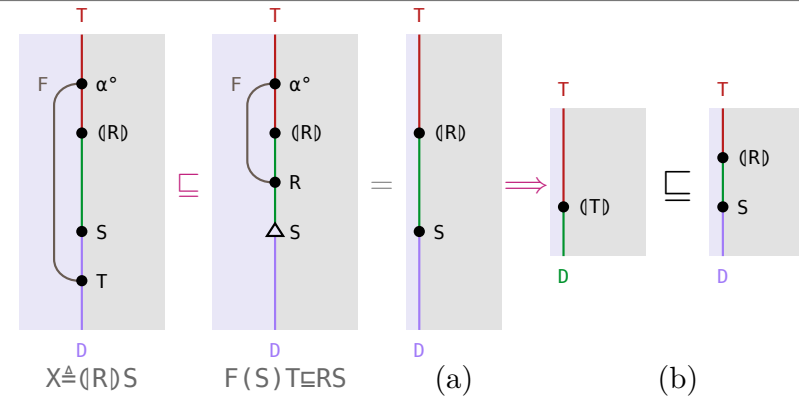
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

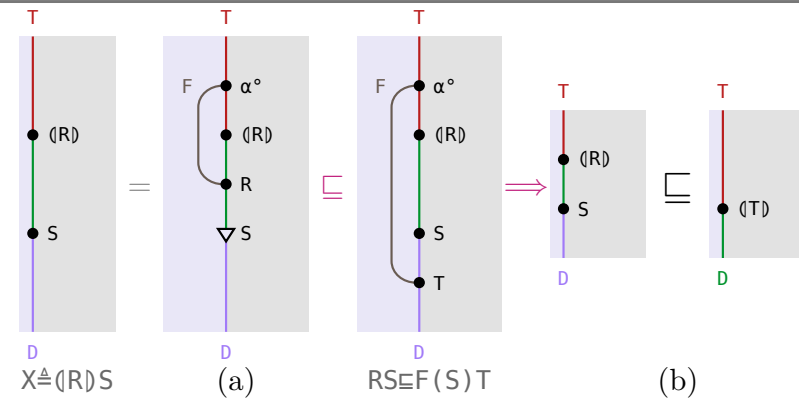


(a)	$\langle R \rangle S$	$F(\langle R \rangle) R = \alpha(\langle R \rangle)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(b)	$\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$	$\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-)T$, and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a)	$\alpha^\circ F(\langle R \rangle) R S$	$\alpha(\langle R \rangle) = F(\langle R \rangle) R$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(b)	$\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$	$\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-)T$, and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$

§6.3. Hylomorphisms

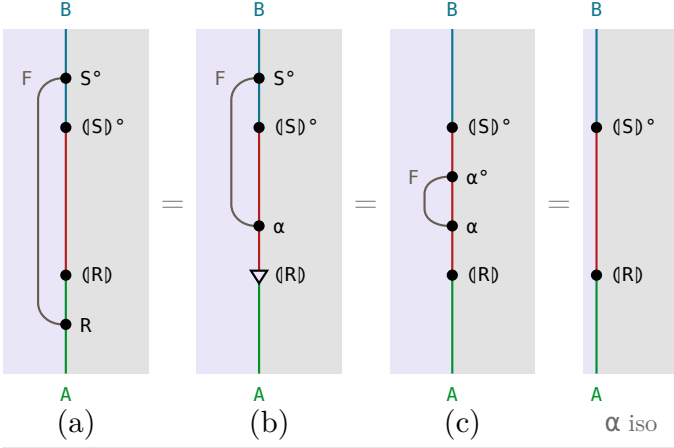
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

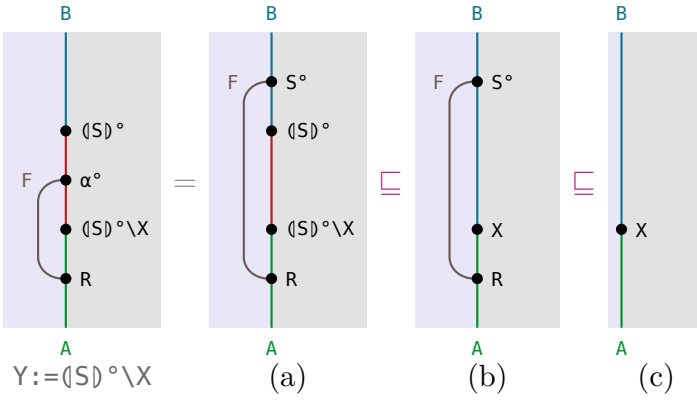
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



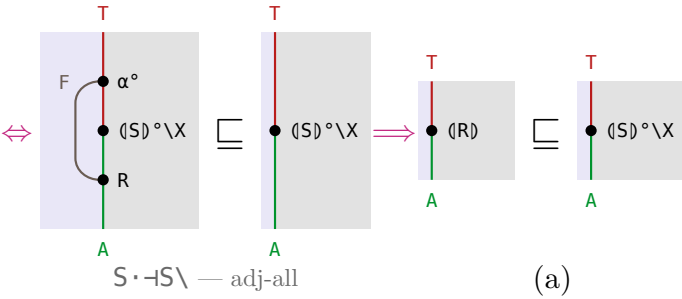
(a)	$S^\circ F((\mathbb{S})^\circ)F((\mathbb{R}))R$	the body at $(\mathbb{S})^\circ(\mathbb{R})$
(b)	$S^\circ F((\mathbb{S})^\circ)\alpha(\mathbb{R})$	$F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq \text{reduce}$
(c)	$(\mathbb{S})^\circ \alpha^\circ \alpha(\mathbb{R})$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$

$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



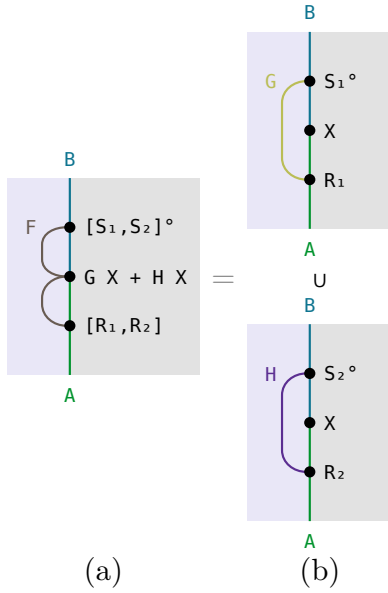
(a)	$S^\circ F((\mathbb{S})^\circ)((\mathbb{S})^\circ \setminus X)R$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$
(b)	$S^\circ F(X)R$	$(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$
(c)	X	$S^\circ F(X)R \sqsubseteq X$



(a)	$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$	$(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.2b), adj-all}$
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$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

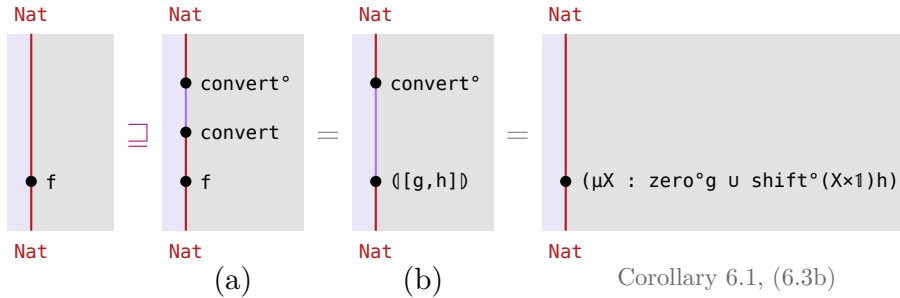


(a)	$[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$	the body of Theorem 6.2 at $F(X)=G(X)+H(X)$
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the $+$ stands for the codiagonals ∇°, ∇ , which are not drawn $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once f sends **zero** to g and turns **shift** into h , the recursion that halves the argument at each step computes f



(a)	$\text{convert}^\circ \text{convert } f$	convert simple
(b)	$\text{convert}^\circ \llbracket [g, h] \rrbracket$	fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes a to the power b

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves a at each step computes the remainder of a by b

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \text{ mod } b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

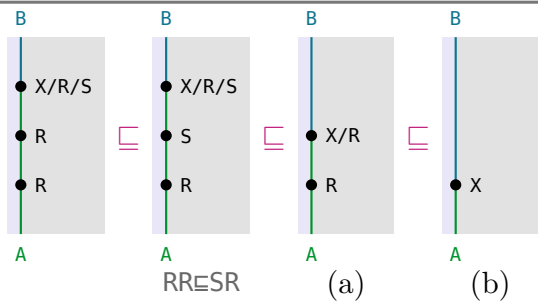
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \in SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $R \in SR$ for an inductive S



(a)	$(X/R)R$	division by S
(b)	X	division by R

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component

$$\text{inductive}(\text{Smember}) \wedge X = SF(X)R \wedge Y = SF(Y)R \Rightarrow X = Y$$

(6.5c)

when S member is inductive, two solutions X, Y of $X = SF(X)R$ are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge SF(X)R \in X \Rightarrow \text{Entire}(X)$$

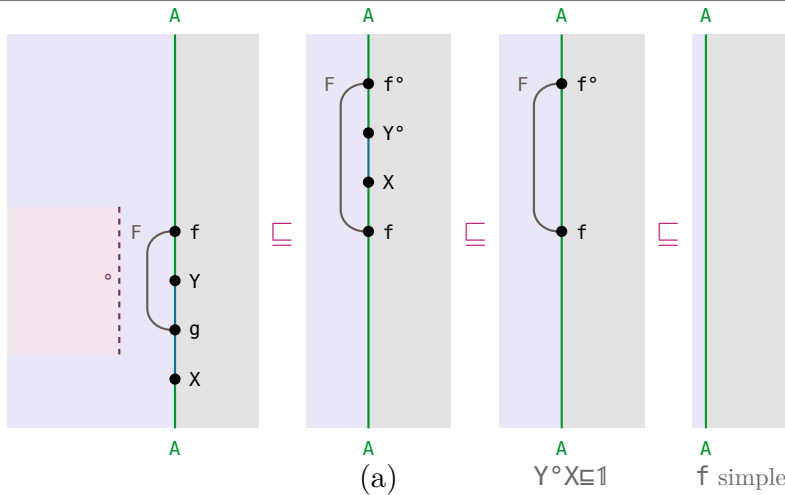
(6.5d)

when S member is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when g member is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

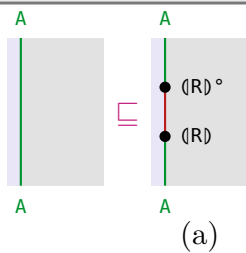


(a)	$f^\circ F(Y^\circ X) f$	$X = gF(X) f, g \text{ simple}$
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$$\text{inductive}(R^\circ \text{member}) \wedge 1 \in R^\circ R \Rightarrow 1 \in (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective

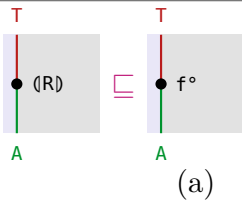


(a)	$(R)^\circ (R)$	(6.5d) at $S = R^\circ$
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$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

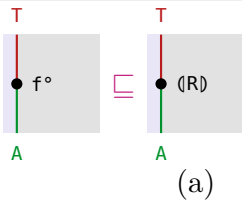
(6.5g)

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)

(a)

 f° shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$ 

(a)

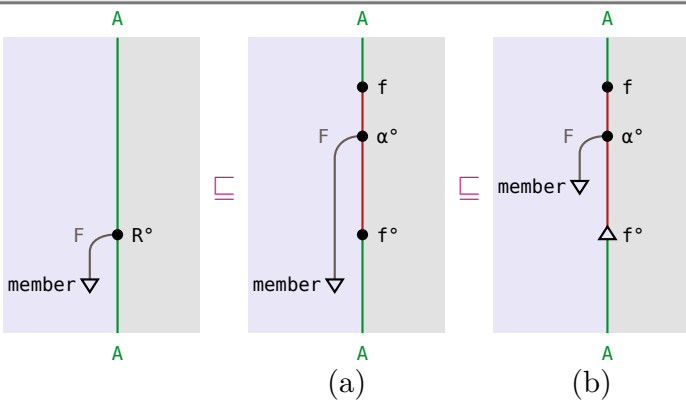
(a)

 $\langle R \rangle$ (6.5f) and $\langle R \rangle f \sqsubseteq 1$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)

(b)

(a)

 $f\alpha^\circ F(f^\circ) \text{member}$ $Rf \sqsubseteq F(f)\alpha$, shunting

(b)

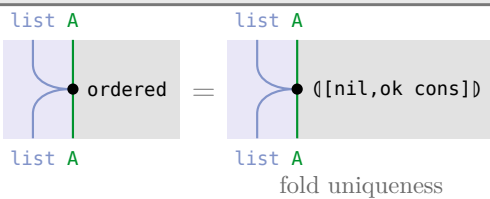
 $f\alpha^\circ \text{member } f^\circ$ member lax natural

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

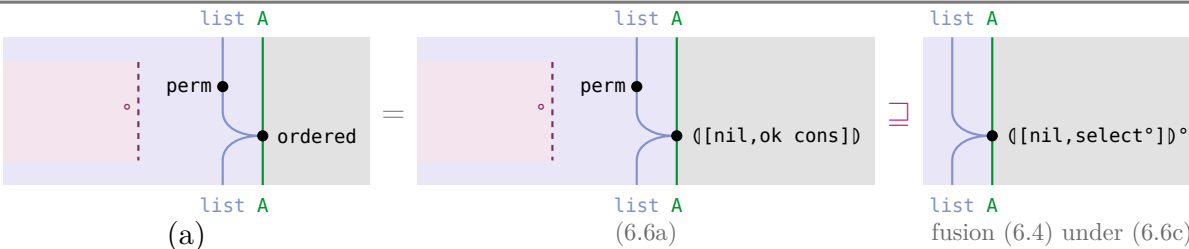


fold uniqueness

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it



(a)

(6.6a)

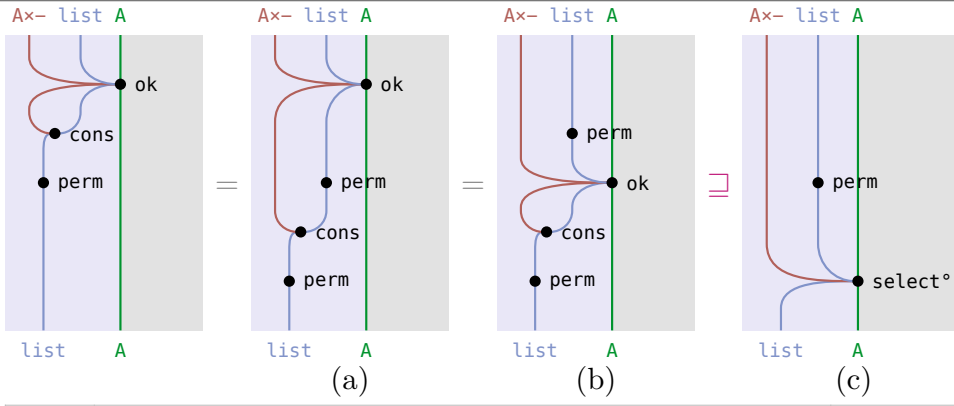
fusion (6.4) under (6.6c)

(a)

 $(\text{ordered perm})^\circ$ $\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

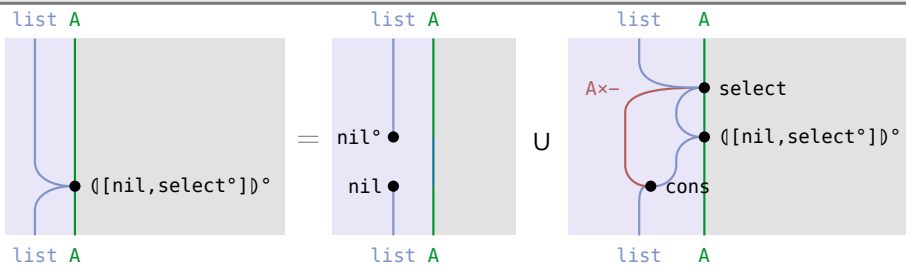
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



(a)	$\text{ok } (1 \times \text{perm}) \text{cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$
(b)	$(1 \times \text{perm}) \text{ok cons perm}$	Ex 6.22, $\text{ok}(1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$
(c)	$(1 \times \text{perm}) \text{select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$

$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on

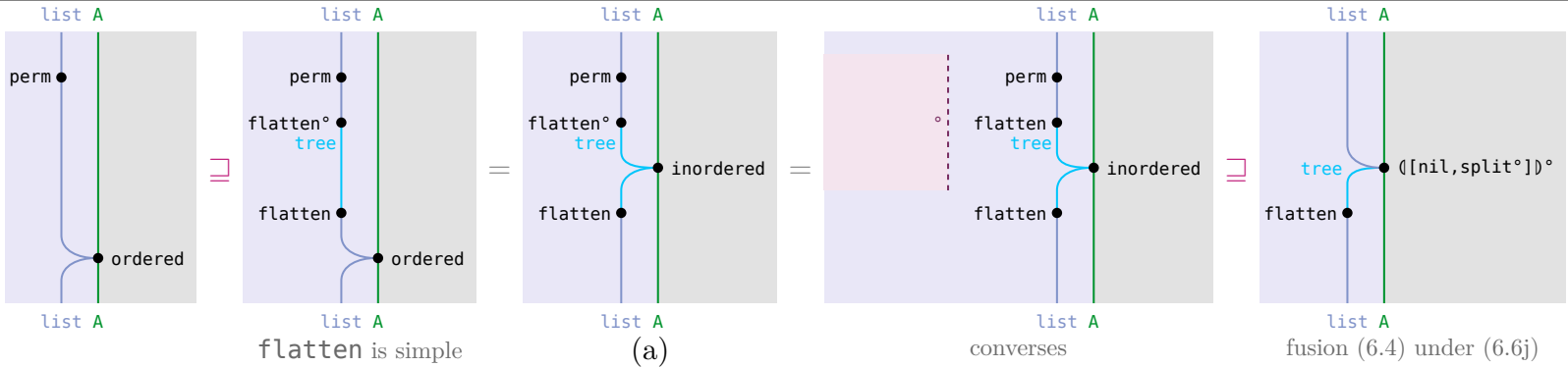


unfolding the converse of a fold

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

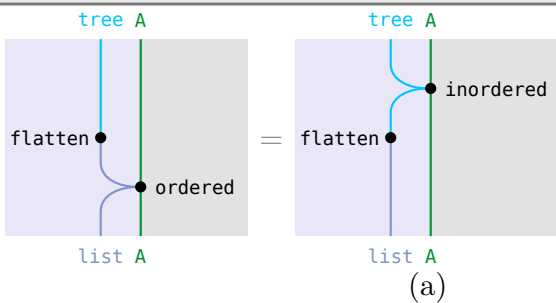
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



(a)	$\text{perm flatten}^\circ \text{inordered flatten}$	$\text{flatten ordered} = \text{inordered flatten}$
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$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

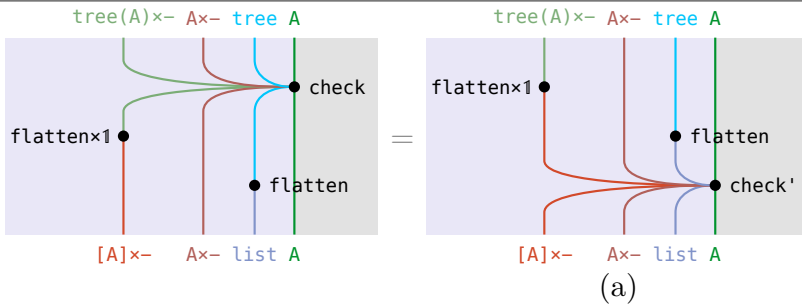


(a)	inordered flatten	R transitive: a flattened tree is ordered iff every node passes check
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check (flatten×(1×flatten))=(flatten×(1×flatten))check'

(6.6g)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



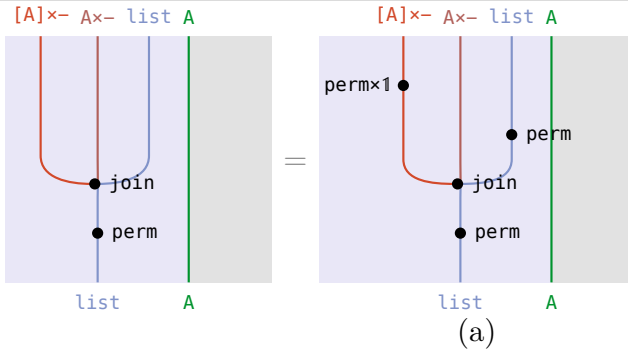
(a) $(\text{flatten} \times (1 \times \text{flatten})) \text{check}'$

$b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$

join perm=(perm×(1×perm))join perm

(6.6h)

permuting the list $x ++ [a] ++ y$ is the same as first permuting x and y and then permuting the joined list



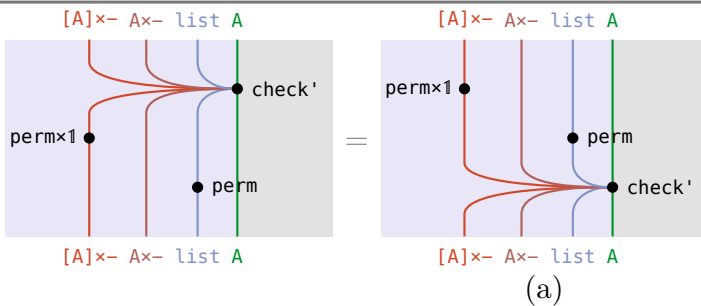
(a) $(\text{perm} \times (1 \times \text{perm})) \text{join perm}$

perm is transitive and closed under $++$

check' (perm×(1×perm))=(perm×(1×perm))check'

(6.6i)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

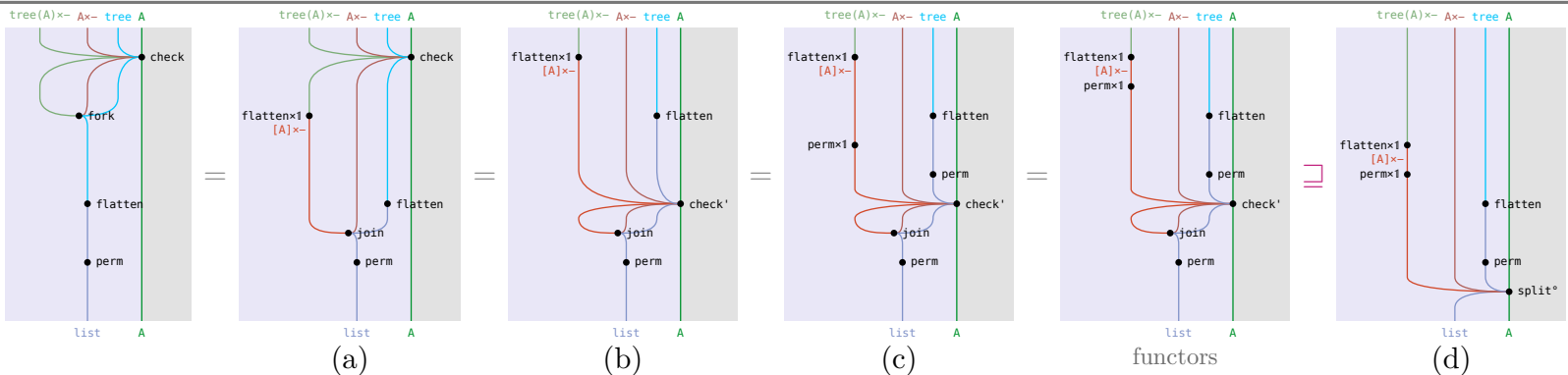


(a) $(\text{perm} \times (1 \times \text{perm})) \text{check}'$

$b \text{ inlist } x \iff b \text{ inlist } \text{perm}(x)$

split° ⊆ check' join perm ⇒ ((flatten perm)×(1×(flatten perm)))split° ⊆ check fork flatten perm (6.6j)

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

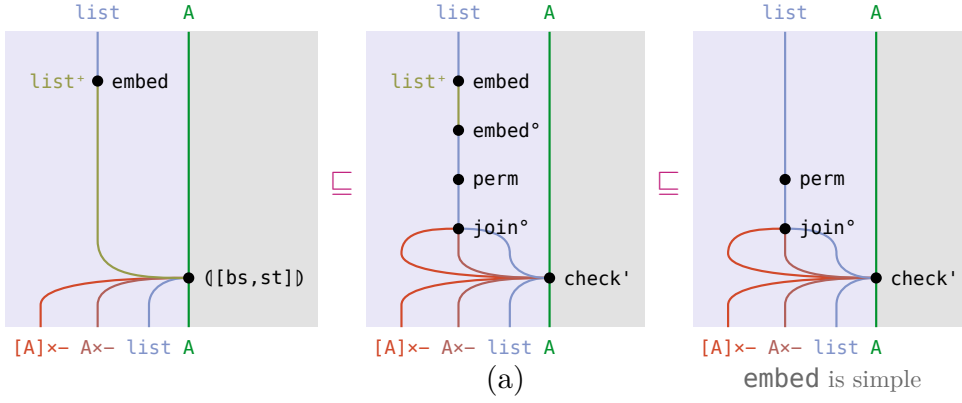


(a)	$\text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{join perm}$	$\text{fork flatten} = F(\text{flatten}) \text{join}$
(b)	$(\text{flatten} \times (1 \times \text{flatten})) \text{check}' \text{join perm}$	$\text{check } F(\text{flatten}) = F(\text{flatten}) \text{check}'$
(c)	$(\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{check}' \text{join perm}$	$\text{join perm} = F(\text{perm}) \text{join perm, check'}$ $F(\text{perm}) = F(\text{perm}) \text{check}'$
(d)	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ$	$\text{split}^\circ \subseteq \text{check}' \text{join perm}$

$$\text{bs}\text{Ewrap perm join}^\circ\text{check}' \wedge (1 \times (\text{perm join}^\circ\text{check}')) \text{stEcons perm join}^\circ\text{check}' \Rightarrow \text{embed}([bs, st]) \text{D}\text{E} \quad (6.6k)$$

perm join[°]check'

turning a list into a non-empty list and folding it with any $[bs, st]$ meeting the two conditions below gives only **checked triples (x, a, y)** whose join $x \# [a] \# y$ is a permutation of the list

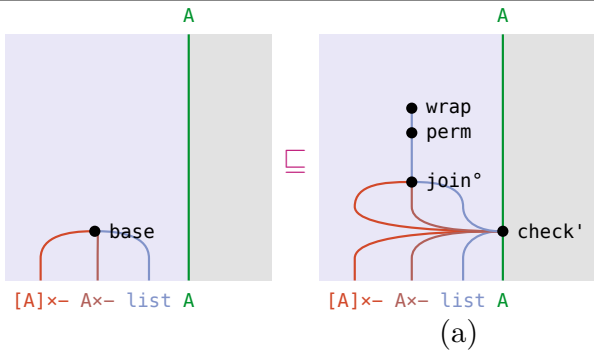


(a) $\text{embed embed}^\circ\text{perm join}^\circ\text{check}'$ (6.6l) and (6.6m), induction on the non-empty list

baseEwrap perm join[°]check'

(6.6l)

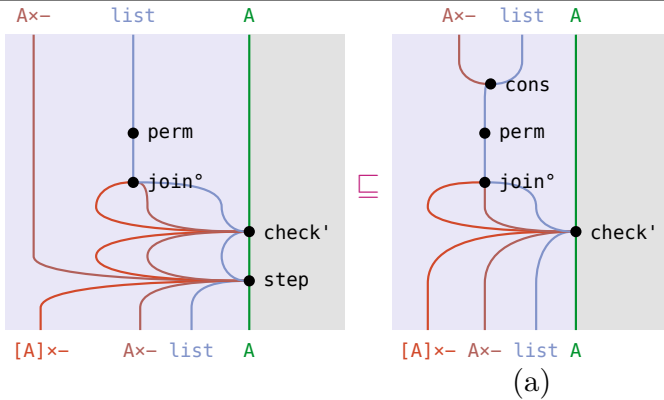
base(a) is a **checked triple** whose join is a permutation of the one-element list $[a]$



(a) $\text{wrap perm join}^\circ\text{check}'$ $[] \# [a] \# [] = [a]$, and check' holds on empty lists

$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ\text{check}')) \text{stepEcons perm join}^\circ\text{check}' \quad (6.6m)$$

if (x, b, y) is a **checked triple** for a permutation of l , then $\text{step}(a, (x, b, y))$ is a **checked triple** for a permutation of $[a] \# l$

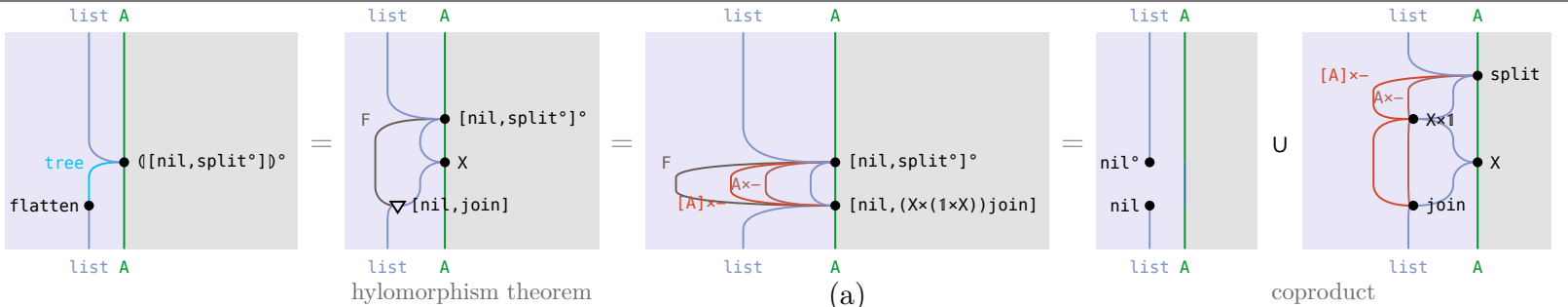


(a) $\text{cons perm join}^\circ\text{check}'$ aRb puts a in front of x ; otherwise bRa and a goes in front of y

$$([nil, \text{split}^\circ])^\circ \text{flatten} = nil^\circ nil \cup \text{split} \quad (6.6n)$$

$$((([nil, \text{split}^\circ])^\circ \text{flatten}) \times (1 \times ([nil, \text{split}^\circ])^\circ \text{flatten})) \text{join}$$

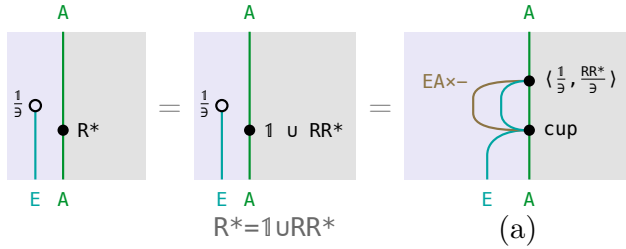
quicksort returns $[]$ on $[]$, and otherwise splits into (x, a, y) , sorts x and y and joins them around a



(a) $[nil, \text{split}^\circ]^\circ [nil, (X \times (1 \times X)) \text{join}]$ $F(X) [nil, \text{join}] = [nil, (X \times 1 \times X) \text{join}]$

$$\frac{R^*}{S} = \left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup} \quad (6.7g)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors



(a)

$$\left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup}$$

$$\Lambda(R \cup S) = (\Lambda(R), \Lambda(S)) \text{cup}$$

$$R - \emptyset = R$$

(6.7h)

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$$

(6.7i)

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

$$SR^* = (\mu X : S \cup XR)$$

(6.7j)

S followed by any number of R steps is the least X containing S and closed under a further R step

$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

$$\theta(\emptyset, S) = SR^*$$

(6.7k)

started with nothing found and S to explore, θ computes SR^*

	(a)	$\theta(\emptyset, S)$	
=	(b)	$\emptyset \cup (\mu X : S \cup (XR - \emptyset))$	definition of θ
=	(c)	$(\mu X : S \cup XR)$	$R - \emptyset = R$ — (6.7h)
=	(d)	SR^*	(6.7j)

$$\theta(P, \emptyset) = P$$

(6.7l)

with nothing left to explore, θ returns what it has found

	(a)	$\theta(P, \emptyset)$	
=	(b)	$P \cup (\mu X : \emptyset \cup (XR - P))$	definition of θ
=	(c)	$P \cup \emptyset$	$\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$
=	(d)	P	$P \cup \emptyset = P$